Induction Machine Short-term Voltage Stability and Protection Measures

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***Abstract*— This paper presents preliminary results on protection measures aiming at minimizing the impact of short-term voltage instability in a part of a power system with large wind power penetration in the form of wind farms equipped with simple induction generators. It is seen that strict protection settings on the induction machines most prone to instability can achieve this goal.**

***Index Terms*— Short-term voltage stability, wind generation, induction machines, overspeed and undervoltage protection.**

I. Introduction

SHORT-term voltage instability is a growing concern for power s systems worldwide [1-3], due to the growing impact of motor load and other fast recovering load devices [4-5]. In this paper we examine a particular case, where short-term instability is caused by large penetration of induction generators connected in a weak area of the grid, where wind power is in abundance [6].

In particular the scope of the paper is to address the problem of setting over-speed and under-voltage protection devices that will minimize the extent of wind power loss in case of a short-term instability. The method of analysis is based on the research results reported in [7] and in particular on the generalized eigenvalue analysis performed during simulation of an unstable scenario, in order to identify the weakest machines, i.e. the ones causing the instability.

Following this analysis, it is found that stricter protection settings on the machines most prone to instability will provide the minimum wind generation shedding. This result is important, as it can be generalized for induction motor protection to provide the minimum load shedding in case of short-term voltage instability.

The paper is organized as follows: Section II describes a short-term voltage instability scenario in a realistic system, while Sections III and IV investigate the effect of undervoltage and over-speed protection settings for wind generators. Section V forms the conclusions of the paper.

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II. Instability of Induction Wind Generators In A Realistic System

A. System Description

The system studied here corresponds roughly to the South Evia region of the Hellenic Interconnected System [6]. The one-line equivalent diagram of the examined system is shown in Fig. 1.

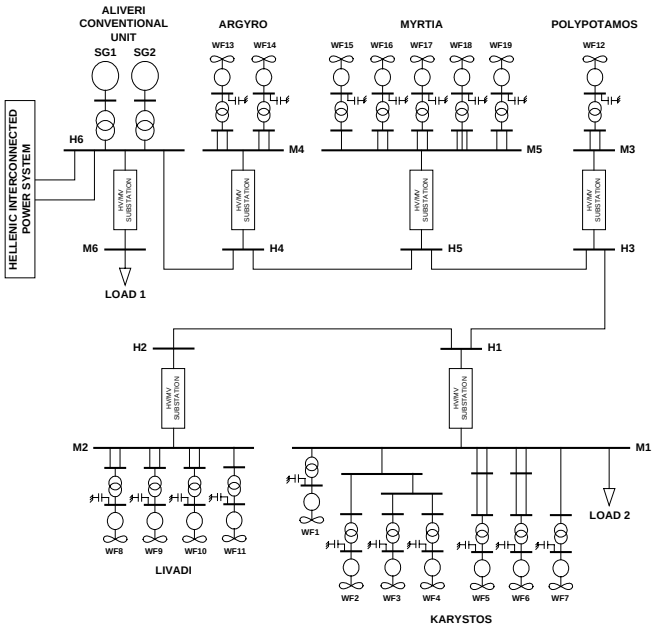


Fig. 1. One-line equivalent diagram of South Evia electrical network.

The examined network consists of a local conventional steam power plant, 19 wind farms of total nominal capacity of 199.5 MW and two aggregate local loads represented as constant impedances. The local generation station consists of two synchronous generators SG1 and SG2 of nominal power 176.5 MVA each. The local generators have AVRs of type AC4 according to IEEE definitions [8], equipped with proportional, summed type OEL with both transient and steady-state field current limitation block [9].

The 19 wind farms (WF1-WF19) are located, as shown in Fig. 1, in the five areas of Argyro, Myrtia, Polypotamos, Karystos and Livadi. Each wind farm is modeled as an equivalent induction generator. The wind farms are connected

to the transmission network (150 kV) through equivalent constant ratio LV/MV transformers, radial MV feeders and HV/MV substations with transformer equipped with on-load tap changers (LTC).

The system interconnection to the main Hellenic Interconnected System is modeled via two equivalent lines, with equivalent impedance corresponding to the short circuit level at the interconnection point.

In the simulation the synchronous generators are represented using a fourth order model. The induction generators are described with a third order model including rotor mechanical and electrical transients. Since the time frame of interest extends to a few seconds, the response of the synchronous generator prime movers and the wind variations are neglected. As a result, the mechanical input torques of both synchronous and induction generators are considered constant during the simulation.

It was seen in [10] that with proper capacitor compensation the system is voltage stable in the long term, even for very severe contingencies. It is important to note that long-term voltage stability is maintained mainly because the LTCs at the HV/MV substations, where wind generators are connected, are able to regulate MV voltage, thus decreasing reactive power consumption by the induction generators. These results led to the conclusion that normal shunt capacitor banks were sufficient to maintain stability by allowing the synchronous machines a wider margin of reactive support and no dynamic reactive compensation was deemed necessary.

In the present paper we consider a slight variant of the above scenario considering the effect of a transient limitation of the generator excitation field. In order to investigate the countermeasures against short-term voltage instability, we deliberately assumed a low value of field current transient limit I_{max}. It should be noted that this does not reflect the degree of security of the actual power system.

Simulation of the system using the WHSSP software developed in NTUA [11] it is seen that the following double contingency leads the system to short-term voltage instability:

- At t=10 s, outage of local generator SG2.
- At t=50 s, loss of one interconnection line.

B. Wind Generator Short-Term Instability

The voltage instability resulting from the transient overexcitation limitation of the remaining synchronous machine is clear from the terminal voltage response of the wind farm WF8 in Livadi shown in Fig. 2-(c). The transient overexcitation limitation of the synchronous machine is shown in Fig. 2-(a).

The short-term voltage instability involves only the induction generators as shown in Figs. 2-(b) and 2-(d) showing the synchronous generator rotor angle and the speed of the equivalent induction generator WF8. Clearly the synchronous machine remains in synchronism while the induction generators successively lose equilibrium and over-accelerate. Thus the disconnection of the wind farms will be the final outcome of instability. The performance of protection settings discussed in this paper is estimated by the total wind generation power loss.

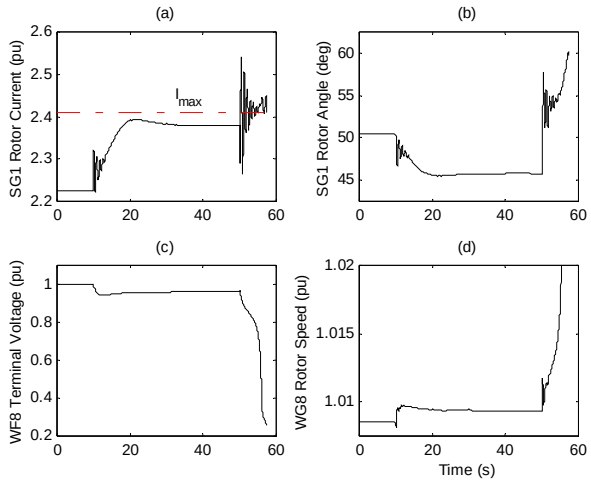


Fig. 2. Simulation results for South Evia network short-term instability: a) field current; b) SG rotor angle; c) WF8 voltage; d) WF8 speed.

C. Eigenvalue Tracking

The above analysis clearly shows that the induction generators constitute the driving force of the short-term instability. The eigenvalue tracking facility is used in order to investigate further which machines contribute mostly in the instability. Fig. 3 shows the evolution of the dominant real eigenvalue for the examined case.

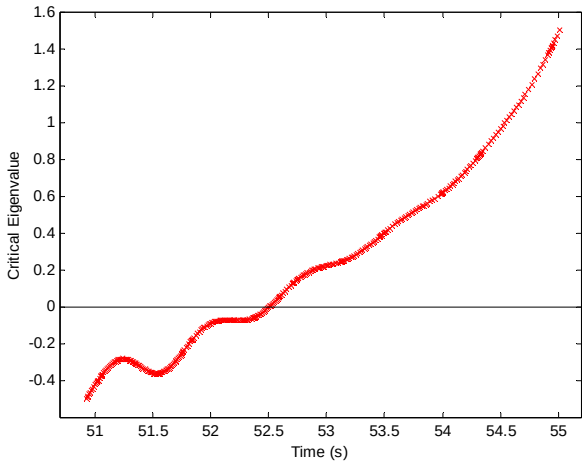


Fig. 3. Evolution of critical eigenvalue.

The dominant eigenvalue corresponds to a monotonic response and appears immediately after the transient overexcitation limitation of the remaining local generator after 50 s. This eigenvalue crosses the origin at t=52.5 s. The dominant state variables, the right eigenvector elements and the participation factors for the zero eigenvalue are shown in Table I. Eigenvector and participation factor analysis shows that the instability involves mainly the three wind farms (WF8-WF10) of Livadi area. The five wind farms (WF1, WF3, WF5-WF7) connected to the next substation of Karystos are also affected, while the other wind farms connected to substations closer to the conventional power plant are affected to a lesser extend.

TABLE I: EIGENVECTOR ELEMENTS AND PARTICIPATION FACTORS FOR ZERO EIGENVALUE

Wind Farm	State Variable	Right Eigenvector		Participation Factor
		Amplitude	*Phase*	
WF1	s	0.0136	0.0	0.158
	E'_d	0.7018	0.0	0.558
WF3	E'_d	0.6958	0.0	0.122
WF5	E'_d	0.7482	0.0	0.264
WF6	s	0.0142	0.0	0.124
	E'_d	0.7654	0.0	0.408
WF7	E'_d	0.7403	0.0	0.340
WF8	s	0.0203	0.0	0.418
	E'_d	**1.0000**	0.0	**1.000**
	E'_q	0.1673	0.0	0.290
WF9	s	0.0197	0.0	0.329
	E'_d	**0.9676**	0.0	**0.814**
	E'_q	0.1518	0.0	0.211
WF10	s	0.0195	0.0	0.320
	E'_d	**0.9544**	0.0	**0.804**
	E'_q	0.1458	0.0	0.199
WF11	s	0.0178	0.0	0.102
	E'_d	**0.8639**	0.0	0.288
WF12	E'_d	0.3808	0.0	0.173
WF13	E'_d	0.3402	0.0	0.115
WF14	E'_d	0.3479	0.0	0.147
WF15	E'_d	0.4430	0.0	0.154
WF16	E'_d	0.4245	0.0	0.235
WF17	E'_d	0.4320	0.0	0.311
WF18	E'_d	0.4232	0.0	0.392
WF19	E'_d	0.4146	0.0	0.182

It is interesting to note that the variable mostly affected by the instability is the flux (transient EMF) rather than the slip of the induction machines. This is due to the fact that the wind generators exhibit large induction machine dynamics [12], i.e. they have an oscillatory response that involves slip and internal EMF angle and a monotonic mode involving magnetic flux. Note also that the critical eigenvalue corresponds to a common mode of response, as the same variables of different wind generators respond in phase [12].

Thus, short-term voltage instability involves mainly the three wind farms (WF8-WF10) of Livadi area, causing their demagnetization and over-acceleration. As long as these machines remain connected to the network, the instability cascades to the nearby wind generators, leading eventually to a local blackout.

III. OVERSPEED WIND GENERATOR SHEDDING

The analysis of instability driving force and mechanism of the previous section III-C is not a mere academic exercise, as it may lead to the correct assessment of possible countermeasures to contain the instability. Since the case has been identified as induction machine instability, the measures to be taken are related to wind generator disconnection.

Thus, the instability can be corrected by overspeed shedding of the wind generators. Note that the overspeed protection relay constitutes an integral part of each wind generator equipment protection, i.e., shaft protection from high wind velocity. In this section we assume that the settings of this relay can be modified (in the direction of becoming stricter), in order to be used as part of a system protection scheme (SPS) that will contain the short-term instability by minimizing the amount of wind power loss.

Initially it is assumed that each wind farm is disconnected from the network, when the rotor speed of the generator exceeds the threshold of 102.5% of synchronous speed for a time interval of 100 ms. The corresponding rotor speed responses of four wind farms are plotted in Fig. 4, where the overspeed threshold is indicated with dashed-dotted lines.

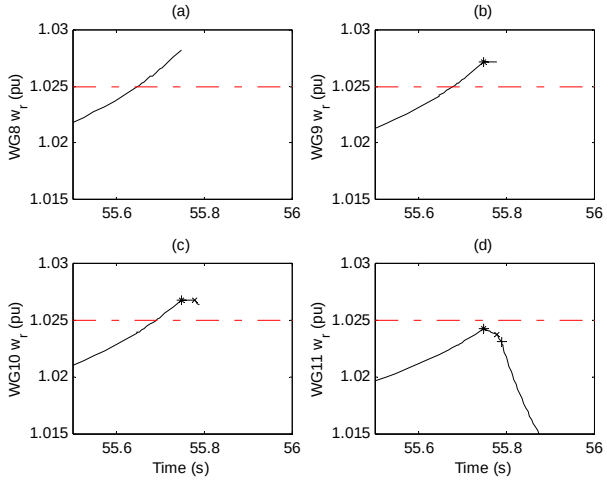


Fig. 4. Wind generator speed of Livadi area ($\Delta\omega_r$>2.5% for 100 ms).

As shown in Fig. 4, the cascaded disconnections of WF8 (marked with "*"), WF9 (marked with "x") and WF10 (marked with "+") lead to the deceleration of WF11. As a result, the latter wind farm remains connected to the network and reaches a post-disturbance equilibrium. The effect of the above overspeed protection scheme results in disconnections of WF8-WF10 of Livadi area, corresponding to loss of 40.2 MW.

The effect of relaxing the above overspeed relay settings are illustrated in Fig. 5, where the rotor speed response of the same wind farms is shown. In this case, the overspeed threshold is increased to 105% of synchronous speed (dashed-dotted lines) for a time interval of 100 ms.

As shown in Fig. 5, the cascade disconnections of WF8 (marked with "*"), WF9 (marked with "x") and WF10 (marked with "+") cannot avoid the over-acceleration of WF11, leading to its subsequent disconnection from the network. With these relaxed overspeed relay settings, eventually seven wind farms are disconnected (all WFs of Livadi area and WF5-WF7 of Karystos), corresponding to loss of 68 MW.

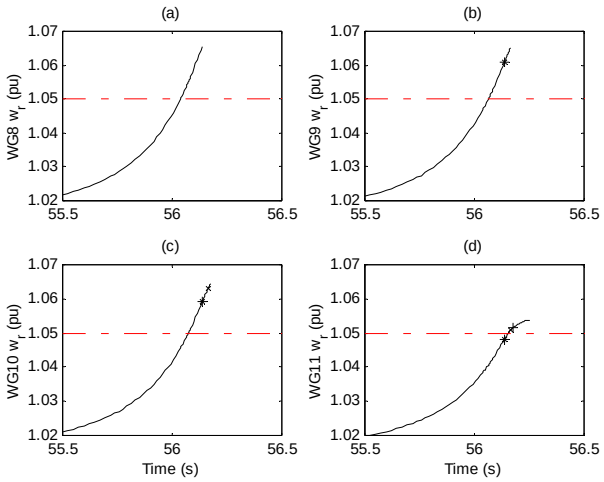


Fig. 5. Wind generator speed of Livadi area ($\Delta\omega_r$>5% for 100 ms).

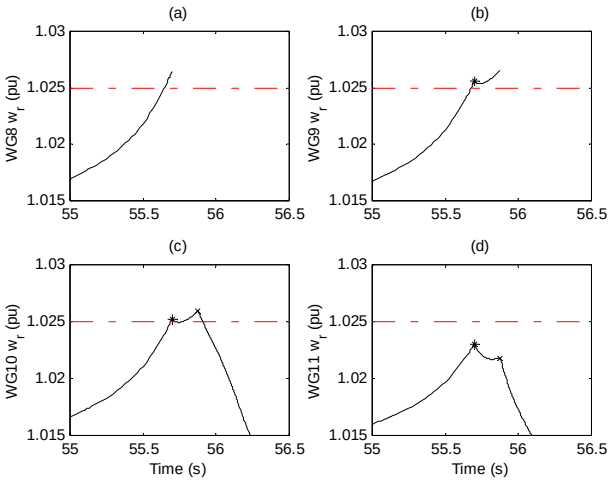


Fig. 6. Wind generator speed of Livadi area (wind farm loss minimization).

Clearly, stricter overspeed protection settings result in losing less wind generation. The reason for this is that by allowing the most affected induction machines to remain connected for a longer time, the evolution of short-term voltage instability will be spread, inducing instability to more machines.

In this respect, it can be noted in Fig. 4 that wind generation loss could be further minimized as WF9 and WF10 were about to start decelerating at the time they were disconnected. Eigenanalysis can be used for the design of more efficient system protection schemes. More specifically, according to the values of Table I, the short-term instability affects mostly the wind farms of Livadi area and especially WF8. By focusing the overspeed protection shedding on this specific area, the following relay settings for WF8-WF11 are proposed:

- The overspeed threshold remains at 102.5% of synchronous speed
- The protection settings for the most affected machine WF8 are made stricter by reducing the time delay for relay action to 50 ms.
- The protection settings for the machines closer to the one most affected WF9-WF11 are relaxed by increasing the delay time to 200 ms.

The rotor speed responses for the above overspeed protection settings are shown in Fig. 6. As seen in this figure, the cascaded disconnection of WF9 (marked with "x") after that of WF8 (marked with "∗") cannot be avoided. However, the subsequent cascading of WF10 and WF11 is avoided as these machines quickly decelerate after the disconnection of WF9 and remain connected to the network reaching a post-disturbance equilibrium. In this case, the loss of wind generation (corresponding to disconnection of WF8 and WF9) is reduced to only 27.6 MW.

Clearly this is an illustrative case only and a more careful design involving also other contingencies should be undertaken for systems with short-term voltage instability problems.

IV. Undervoltage Wind Generator Shedding

A second countermeasure against short-term instability is the undervoltage shedding of induction machines.

Figure 7 shows the terminal voltage response of the wind farms in Livadi area, assuming that each wind farm is disconnected from the network, if its terminal voltage drops below 0.9 pu (dashed-dotted lines) for a time interval of 2.5 s.

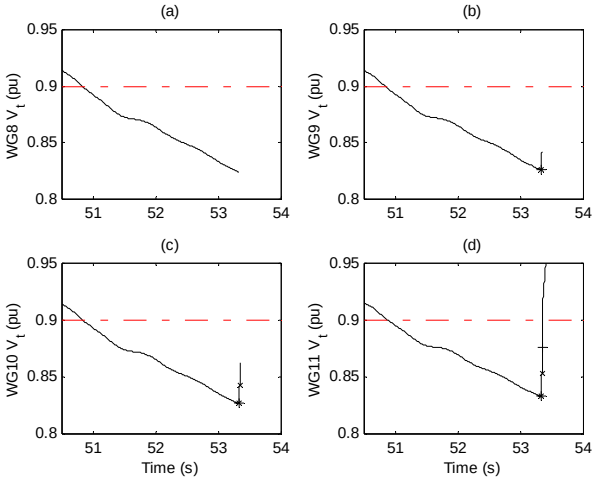


Fig. 7. Wind generator terminal voltage of Livadi area (V_t<0.9 αμ for 2.5 s).

As shown in Fig. 7, the successive disconnection of WF8 (marked with "∗"), WF9 (marked with "x") and WF10 (marked with "+") lead to the restoration of WF11 terminal voltage. As a result, the latter machine decelerates to a post-disturbance stable equilibrium and remains connected to the network. In this case, the total loss of wind generation corresponds to 40.2 MW.

Assuming more permissive protection settings, i.e., the same undervoltage threshold of 0.9 pu (dashed-dotted lines), but a longer time delay of 5 s, the wind farm responses are those of Fig. 8.

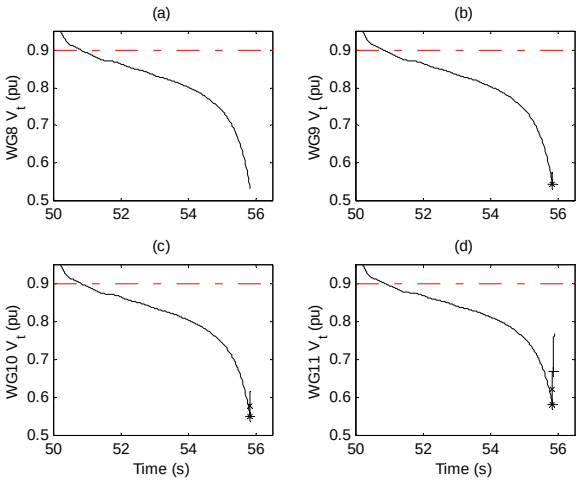


Fig. 8. Wind generator terminal voltage of Livadi area (V_t<0.9 αμ for 5 s).

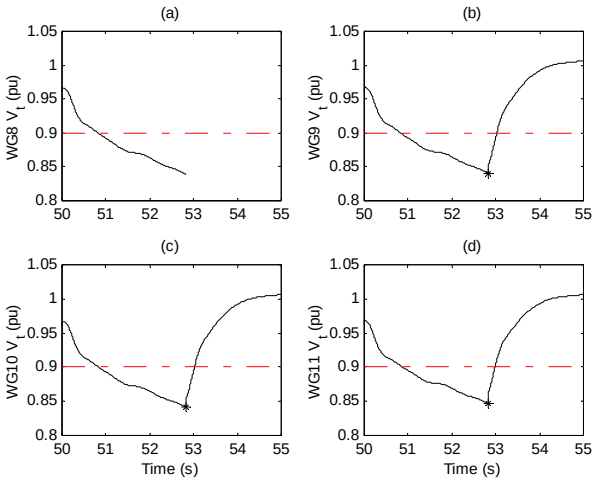


Fig. 9. Wind generator terminal voltage of Livadi (minimum wind farm loss).

In this case, the delayed disconnection of WF8 (marked with "∗"), WF9 (marked with "x") and WF10 (marked with "+") lead to the cascaded disconnection of WF11, even though the latter's terminal voltage was in the process of getting restored. Eventually, for the examined undervoltage relay settings, eight wind farms are disconnected (all WFs of Livadi area and WF1, WF5-WF7 of Karystos area), corresponding to a total loss of 81.5 MW.

Thus, the general rule that stricter protection results in reduced loss of equipment applies also for undervoltage protection relay settings.

As in the overspeed relay settings, a further search to minimize the loss of wind generation can be undertaken. This leads to the following scheme:, it is assumed that only the area are equipped with with the following settings:

- Undervoltage protection relays of all wind farms except those of Livadi are set to a low threshold (e.g. 0.8 pu).
- The undervoltage threshold for Livadi WFs remains the same as before (0.9 pu).
- The time delay for WF8 is set to 2 s (stricter protection).
- The time delay for WF9-WF11 is set to 3 s (more permissive protection).

The terminal voltage of the wind farms with the above undervoltage protection settings is shown in Fig. 9. In this case, as shown in Fig. 9, the disconnection of WF8 (marked with "∗") is sufficient for the terminal voltage restoration for all the remaining wind farms of Livadi area and the rest of the system. As a result, the loss of wind generation is limited to the minimum of 15 MW.

V. Conclusion

This paper presented an illustrative case of short-term voltage instability due to induction generators in a windy area of the Hellenic Power System with weak local transmission and suggested indicative settings for system protection.

Both overspeed and undervoltage relays were assumed and their settings were determined with the objective to minimize the amount of wind generation loss after a specific severe contingency. In fact, it was possible by properly selecting undervoltage relay settings to limit the wind generator loss to only one wind farm, that mostly affected by the instability.

The induction machines mostly affected by the instability were identified using eigenvalue and eigenvector analysis and an on-line linearization performed during the simulation.

The general rule suggested is that stricter protection settings for the machines more prone to instability can minimize the extent of machine disconnections and avoid cascaded trips.

Of course, in actual cases of short-term voltage instability, a more thorough investigation, including the simulation of other contingencies and consideration of other protection goals (e.g. to avoid islanding) is necessary. However this paper hopefully serves as an illustration of general trends that could help in the design of system protection schemes against short-term voltage instability

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